

Practical Lab Task: Parcel Delivery Database Development using phpMyAdmin

Duration: 2 Hours

Course

Database Management Systems / Information Systems Laboratory

Practical Task Title

Development of a Parcel Delivery Database System using phpMyAdmin

1. Task Overview

Students are required to develop a simple relational database for a parcel delivery company using **phpMyAdmin** and **MySQL** within **TWO (2) HOURS**.

The task focuses on:

- Creating a database
- Designing relational tables
- Establishing primary and foreign keys
- Inserting sample data
- Executing SQL queries

Students must create **EIGHT (8) related tables** based on the given scenario.

2. Scenario

A parcel delivery company named **CepatSampai Delivery Services** manages parcel deliveries across several cities in Malaysia.

The company currently faces several operational issues because information is stored manually:

- Parcel information is difficult to track
- Customer records are duplicated
- Delivery assignments are not properly monitored
- Payment records are inconsistent
- Branch managers cannot generate quick reports

To improve efficiency, the company wants a simple database system developed using phpMyAdmin.

As a junior database developer, you are required to build the database structure for the company.

3. Database Requirements

Students must create a database named:

parcel_delivery_db

The database must contain the following **8 tables**:

No	Table Name	Purpose
1	customers	Store customer information
2	receivers	Store receiver information
3	branches	Store branch information
4	delivery_staff	Store delivery staff information
5	parcels	Store parcel details
6	tracking	Store parcel tracking records
7	payments	Store payment information
8	delivery_assignment	Store parcel delivery assignments

Table	Action	Rows	Type	Collation	Size	Overhead
☐ branches	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ customers	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ delivery_assignment	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ delivery_staff	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ parcels	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ payments	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ receivers	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ tracking	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
8 tables	Sum	0	InnoDB	utf8mb4_general_ci	240.0 KiB	0 B

4. Minimum Attributes Required

Students may add additional attributes if necessary.

Table 1: customers

Attribute	Data Type
customer_id	INT
customer_name	VARCHAR(100)
phone_number	VARCHAR(20)
email	VARCHAR(100)

	Browse	Structure	SQL	Search	Insert	Export	Import	Privileges	Operations	Tracking
	Table structure									
	Relation view									
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action	
☐ 1	customer_id	int(11)			No	None		AUTO_INCREMENT	Change	Drop More
☐ 2	customer_name	varchar(100)	utf8mb4_general_ci		No	None			Change	Drop More
☐ 3	phone_number	varchar(20)	utf8mb4_general_ci		No	None			Change	Drop More
☐ 4	email	varchar(100)	utf8mb4_general_ci		Yes	NULL			Change	Drop More

Table 2: receivers

Attribute	Data Type
receiver_id	INT
receiver_name	VARCHAR(100)
phone_number	VARCHAR(20)
address	VARCHAR(200)

Server: 127.0.0.1 » Database: parcel_delivery_db » Table: receivers										
	Browse	Structure	SQL	Search	Insert	Export	Import	Privileges	Operations	Track
	Table structure									
	Relation view									
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action	
☐ 1	receiver_id	int(11)			No	None		AUTO_INCREMENT	Change	Drop More
☐ 2	receiver_name	varchar(100)	utf8mb4_general_ci		No	None			Change	Drop More
☐ 3	phone_number	varchar(20)	utf8mb4_general_ci		No	None			Change	Drop More
☐ 4	address	varchar(200)	utf8mb4_general_ci		No	None			Change	Drop More

Table 3: branches

Attribute	Data Type
branch_id	INT
branch_name	VARCHAR(100)
city	VARCHAR(50)
contact_number	VARCHAR(20)

Server: 127.0.0.1 » Database: parcel_delivery_db » Table: branches									
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking									
Table structure Relation view									
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	branch_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	branch_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	city	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐ 4	contact_number	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More

Table 4: delivery_staff

Attribute	Data Type
staff_id	INT
staff_name	VARCHAR(100)
phone_number	VARCHAR(20)
branch_id	INT

Server: 127.0.0.1 » Database: parcel_delivery_db » Table: delivery_staff									
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking									
Table structure Relation view									
#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	staff_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	staff_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	phone_number	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐ 4	branch_id	int(11)			No	None			Change Drop More

Table 5: parcels

Attribute	Data Type
parcel_id	INT
customer_id	INT
receiver_id	INT
parcel_weight	DECIMAL(5,2)
delivery_type	VARCHAR(50)
delivery_fee	DECIMAL(6,2)

The screenshot shows a database management interface with a top navigation bar containing icons for Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, and Tracking. Below the navigation bar, there are tabs for 'Table structure' and 'Relation view'. The 'Table structure' tab is active, displaying a table with 6 columns: #, Name, Type, Collation, Attributes, Null, Default, Comments, Extra, and Action. The table lists the following attributes:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	parcel_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	customer_id	int(11)			No	None			Change Drop More
3	receiver_id	int(11)			No	None			Change Drop More
4	parcel_weight	decimal(5,2)			No	None			Change Drop More
5	delivery_type	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
6	delivery_fee	decimal(6,2)			No	None			Change Drop More

Below the table, there are several controls: a 'Check all' checkbox, a 'With selected:' dropdown, and buttons for 'Browse', 'Change', 'Drop', 'Primary', 'Unique', 'Index', 'Spatial', and 'Fulltext'. At the bottom, there are buttons for 'Add to central columns' and 'Remove from central columns'.

Table 6: tracking

Attribute	Data Type
tracking_id	INT
parcel_id	INT
tracking_status	VARCHAR(100)
update_date	DATE

The screenshot shows a database management interface with a top navigation bar containing icons for Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, and Tracking. Below the navigation bar, there are tabs for 'Table structure' and 'Relation view'. The 'Table structure' tab is active, displaying a table with 4 columns: #, Name, Type, Collation, Attributes, Null, Default, Comments, Extra, and Action. The table lists the following attributes:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	tracking_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	parcel_id	int(11)			No	None			Change Drop More
3	tracking_status	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
4	update_date	date			No	None			Change Drop More

Table 7: payments

Attribute	Data Type
payment_id	INT
parcel_id	INT
payment_method	VARCHAR(50)
amount_paid	DECIMAL(6,2)

The screenshot shows a database management interface with the following components:

- Top bar: Server: 127.0.0.1 » Database: parcel_delivery_db » Table: payments
- Navigation tabs: Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, Tracking
- Sub-tabs: Table structure (selected), Relation view
- Table structure table:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 payment_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 parcel_id	int(11)			No	None			Change Drop More
☐	3 payment_method	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 amount_paid	decimal(6,2)			No	None			Change Drop More

Table 8: delivery_assignment

Attribute	Data Type
assignment_id	INT
parcel_id	INT
staff_id	INT
delivery_status	VARCHAR(50)

The screenshot shows a database management interface with the following components:

- Top bar: Server: 127.0.0.1 » Database: parcel_delivery_db » Table: delivery_assignment
- Navigation tabs: Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, Operations, Tracking
- Sub-tabs: Table structure (selected), Relation view
- Table structure table:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 assignment_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 parcel_id	int(11)			No	None			Change Drop More
☐	3 staff_id	int(11)			No	None			Change Drop More
☐	4 delivery_status	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More

5. Relationship Requirements

Students must establish the following relationships:

Parent Table	Child Table
customers	parcels
receivers	parcels
branches	delivery_staff
parcels	tracking
parcels	payments
parcels	delivery_assignment
delivery_staff	delivery_assignment

